

Recall — the circle and the disc

The curve and the region it encloses are two different things, with two different names.

LESSON 89

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 8.2 ENTRY

LESSON CLOCK

10'

12'

12'

6'

Circle and disc	10 min
The parts	12 min
Radius and diameter	12 min
Drawing one	6 min

LEARNING OBJECTIVES

- Distinguish a circle from a disc.
- Name the centre, radius, diameter, chord and arc.
- Draw a circle of a given radius with compasses.
- State the relation between radius and diameter.

TEXTBOOK

National	pp. 250–253
Cambridge	Stage 7, pp. 80–84
Workbook	Exercise 8.1

01

Explanation

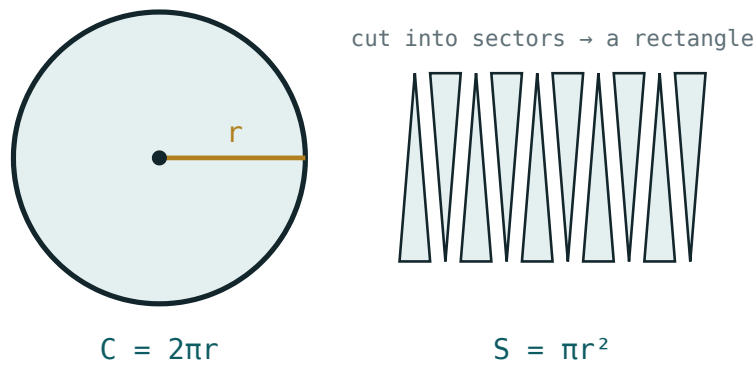
Circle and disc

WORD	MEANS	MEASURED BY
circle	the curve itself	a length — the circumference
disc	the curve and everything inside	an area

A CIRCLE IS A SET OF POINTS AT ONE DISTANCE FROM THE CENTRE

Every point of the curve is exactly r from the centre — no nearer, no further. That single sentence is the definition, and the compasses are its machine.

The parts



The centre, radius, diameter and chord

PART	WHAT IT IS
centre	the fixed point
radius	from the centre to the curve
diameter	right across, through the centre
chord	joining two points of the curve
arc	part of the curve
sector	a slice, between two radii
segment	the piece cut off by a chord

A DIAMETER IS A CHORD; A CHORD IS USUALLY NOT A DIAMETER

The diameter is the longest chord — the only one passing through the centre. Every other chord is shorter.

Radius and diameter

$$d = 2r \quad r = \frac{d}{2}$$

R	D
3 cm	6 cm
7 cm	14 cm
4.5 cm	9 cm
12 cm	24 cm

READ WHICH ONE THE QUESTION GIVES

A circle “of diameter 14 ” has radius 7 . Half the errors in the next three lessons come from using the wrong one of the two in a formula.

Drawing one

STEP	WHAT TO DO
1	open the compasses to the radius against a ruler
2	put the point on the centre and mark it
3	turn the compasses in one smooth movement
4	label the centre, and mark a radius and a diameter

SET THE COMPASSES TO THE RADIUS, NEVER THE DIAMETER

For a circle of diameter 10 cm the compasses open to 5 cm . The commonest practical error in this topic is drawing a circle twice the size wanted.

02 Worked examples**EXAMPLE 1** A circle has diameter *14 cm*. Find its radius.

$$① \quad r = \frac{d}{2}$$

$$② \quad = 7 \text{ cm}$$

ANSWER*7 cm***EXAMPLE 2** What is the difference between a circle and a disc?

① A circle is the curve alone.

② A disc is the curve together with the region inside it.

③ One is measured by a length, the other by an area.

ANSWER

Curve against region

EXAMPLE 3 Is every chord a diameter? Is every diameter a chord?

- ① A chord joins two points of the curve.
- ② A diameter does that and passes through the centre.
- ③ So every diameter is a chord; most chords are not diameters.

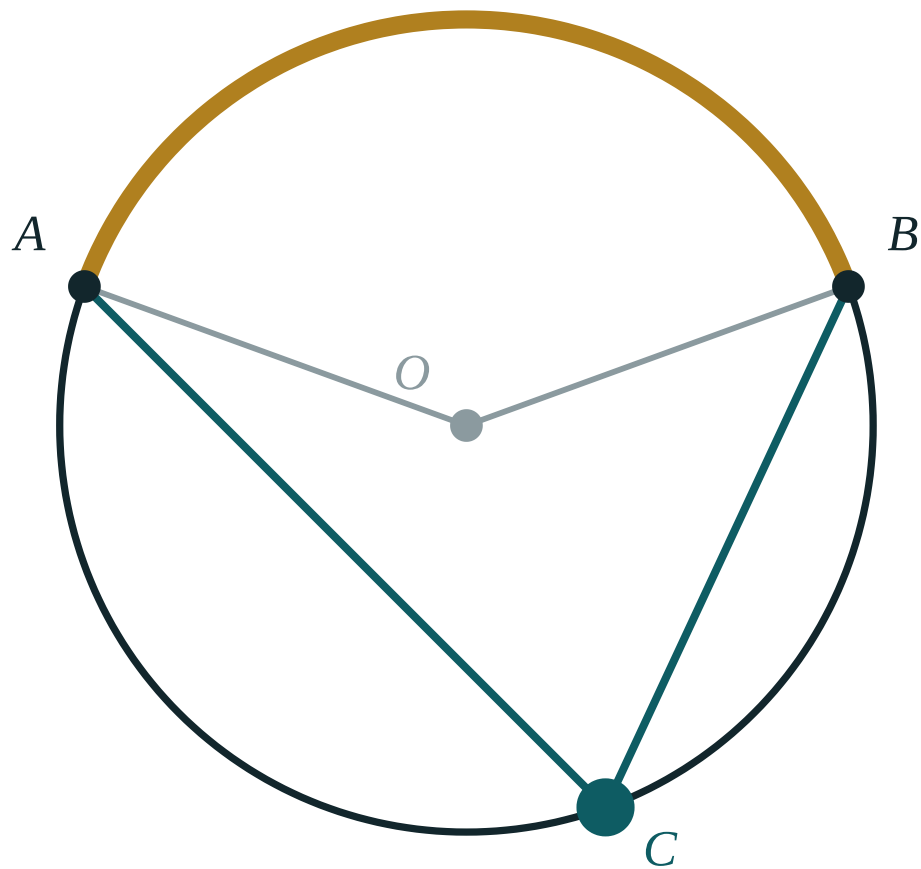
ANSWER

No; yes

03 **Interactive model**

Ask the class to draw a circle of diameter 8 cm; the ones that come out 16 cm across make the radius-diameter point better than any warning.

The parts of a circle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Circle	Aylana	Окружность
Disc	Doira	Круг
Centre	Markaz	Центр
Radius	Radius	Радиус
Diameter	Diametr	Диаметр
Chord	Vatar	Хорда
Arc	Yoy	Дуга
Compasses	Sirkul	Циркуль

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A circle is:

the curve

the region inside

both

neither

A disc is measured by:

a length

an area

an angle

a radius only

The diameter is:

r

$2r$

$\frac{r}{2}$

πr

A circle of diameter *14* has radius:

28

7

14

3.5

The longest chord is:

a radius

the diameter

an arc

a sector

Compasses are set to:

the diameter

the radius

the circumference

any length

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The curve itself is called

ANSWER a circle

2. The region inside is called

ANSWER a disc

3. $r = 3$: the diameter

ANSWER 6 cm

4. $d = 14$: the radius

ANSWER 7 cm

5. $r = 4.5$: the diameter

ANSWER 9 cm

6. A line from the centre to the curve

ANSWER a radius

7. A line joining two points of the curve

ANSWER a chord

Medium · the standard question

7 PROBLEMS

1. The longest chord in a circle

ANSWER the diameter

2. Part of the curve

ANSWER an arc

3. A slice between two radii

ANSWER a sector

4. The piece cut off by a chord

ANSWER a segment

5. Is every diameter a chord?

ANSWER Yes

6. Is every chord a diameter?

ANSWER No

7. Compasses for a circle of diameter 10 cm open to

ANSWER 5 cm

Hard · stretch and reason

7 PROBLEMS

1. A circle of radius r : the longest chord

ANSWER $2r$

2. Two circles of radius 5 with centres 8 cm apart: do they cross?

ANSWER Yes — $8 < 10$

3. Two circles of radius 5 with centres 12 cm apart

ANSWER No — they are apart

4. A chord 6 cm from the centre of a circle of radius 5 cm

ANSWER Impossible — no such chord

5. How many diameters has a circle?

ANSWER Infinitely many

6. Where do all the diameters meet?

ANSWER At the centre

7. Why is a circle drawn with compasses?

ANSWER They keep the distance from the centre fixed

07 Homework

Homework — 5 tasks

Set the compasses to the radius; halve the diameter first if that is what you are given.

1. Draw circles of radius 3 cm and of diameter 9 cm.
2. Mark and label a radius, a diameter, a chord and an arc on one of them.
3. Find the radius of circles with diameters 18 cm and 7 cm.
4. Find the diameter of circles with radii 6.5 cm and 11 cm.
5. Write one sentence explaining the difference between a circle and a disc.